

Exploratory Data Analysis: Player Similarity Feature Exploration

Project Goal

This notebook performs exploratory data analysis on SkillCorner tracking data and StatsBomb event data to identify discriminative features for computing player similarity scores. The ultimate goal is to build a system that identifies the best alternative player for any given player based on movement patterns, spatial behavior, and action profiles.

Notebook Structure

- 1. **Data Overview** - Dataset descriptions, scope, and structure
- 2. **Movement Intensity Profiles (SkillCorner)** - Speed metrics and movement patterns
- 3. **Spatial Footprint & Positional Behavior (SkillCorner)** - Position distributions and zone occupancy
- 4. **Action Profiles (StatsBomb)** - Event-based behavioral features
- 5. **Interaction Signal (SkillCorner)** - Relational spacing metrics

Section 1: Data Overview

Dataset Descriptions

SkillCorner Tracking Data: Continuous positional data capturing the (x, y) coordinates of all 22 players and the ball at 10 frames per second. Data is derived from broadcast video using computer vision, with positions extrapolated when players are off-screen. The coordinate system uses meters with the pitch center as origin.

StatsBomb Event Data: Discrete event-by-event data recording on-ball actions (passes, shots, duels, etc.) with associated locations, outcomes, and contextual attributes. Events are manually annotated and include tactical context such as formations and play patterns. The coordinate system uses a 120x80 unit grid with origin at the bottom-left corner.

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DATA OVERVIEW

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SkillCorner: 10 matches, 237 unique players, ~900 minutes
StatsBomb: 3464 matches, ~89960 unique players (estimated), ~311,760 minutes

	Dataset	Matches	Players (est.)	Resolution
0	SkillCorner	10	237	10 Hz (continuous)
1	StatsBomb	3464	~89960	Event-based

Context Summary

The SkillCorner dataset provides dense spatiotemporal tracking data from 10 A-League matches, enabling analysis of continuous movement patterns and positional behavior. The StatsBomb dataset offers extensive event-level coverage across 3,464 matches from multiple competitions, providing rich behavioral profiles based on on-ball actions.

For player similarity, SkillCorner data enables features related to movement intensity, spatial footprint, and relational positioning. StatsBomb data enables features related to action profiles, passing behavior, and event-based signatures. The combination of both datasets supports a comprehensive multi-modal approach to player representation.

Section 2: Movement Intensity Profiles (SkillCorner)

This section analyzes movement intensity features from SkillCorner tracking data to understand how players differ in their physical activity patterns. These features are highly discriminative for player similarity as they capture fundamental differences in playing style and physical demands.

Key Terms and Definitions

Mean Speed: Average movement speed across all frames. *Computation:* For each frame, speed is calculated as distance traveled between consecutive frames divided by time interval (0.1 seconds at 10 Hz). Mean speed = `np.mean(speeds)` where `speeds` is the array of all instantaneous speeds.

90th Percentile Speed (P90 Speed): The speed value below which 90% of a player's speed measurements fall. This metric captures high-intensity movement capacity, representing the player's typical maximum effort speed rather than rare peak values. *Computation:* `np.percentile(speeds, 90)`.

Sprint Threshold: A speed of 7 m/s (25.2 km/h) used to classify high-intensity running. Any frame where a player's speed exceeds this threshold is counted as a sprint.

Sprint Frequency (per 90 min): Number of sprint frames normalized to a 90-minute match duration. *Computation:* `(sprint_count / total_frames) * frames_per_90`, where `sprint_count = np.sum(speeds > 7.0)`, `total_frames` is the number of frames with valid speed data, and `frames_per_90 = 54000` (10 Hz × 60 seconds × 90 minutes). This normalization allows fair comparison across players with different playing times.

Correlation Coefficient (r): A measure of linear relationship strength between two variables, ranging from -1 (perfect negative correlation) to +1 (perfect positive correlation). Values near 0 indicate weak or no linear relationship. *Computation:* Pearson correlation coefficient using `pd.DataFrame.corr()`.

Features analyzed:

- Mean speed (m/s)
- 90th percentile speed (m/s)
- Sprint frequency (per 90 min)

Key questions:

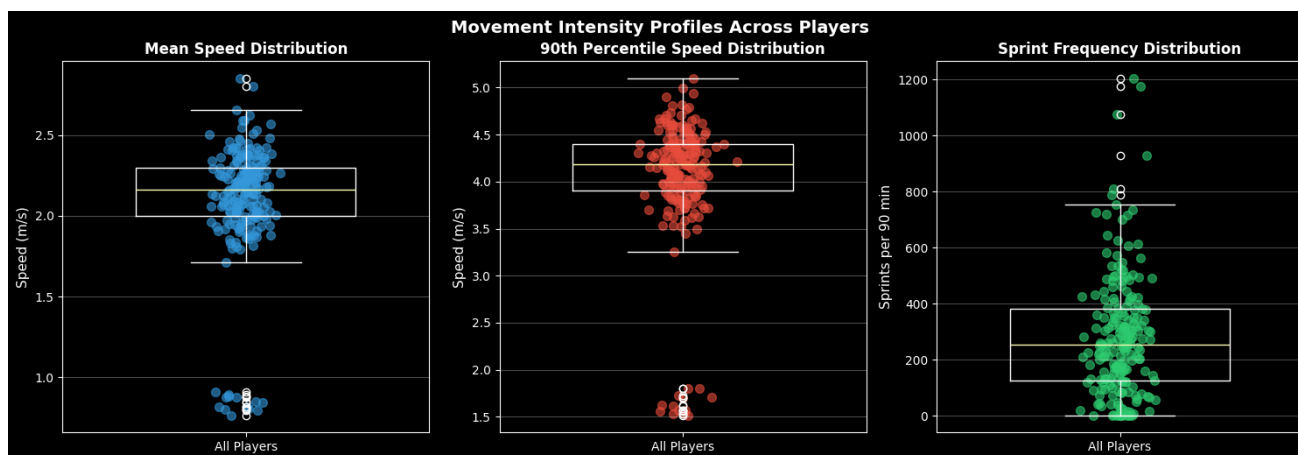
1. Do players differ meaningfully in movement intensity?
2. Are speed metrics redundant, or do they capture distinct aspects of movement behavior?

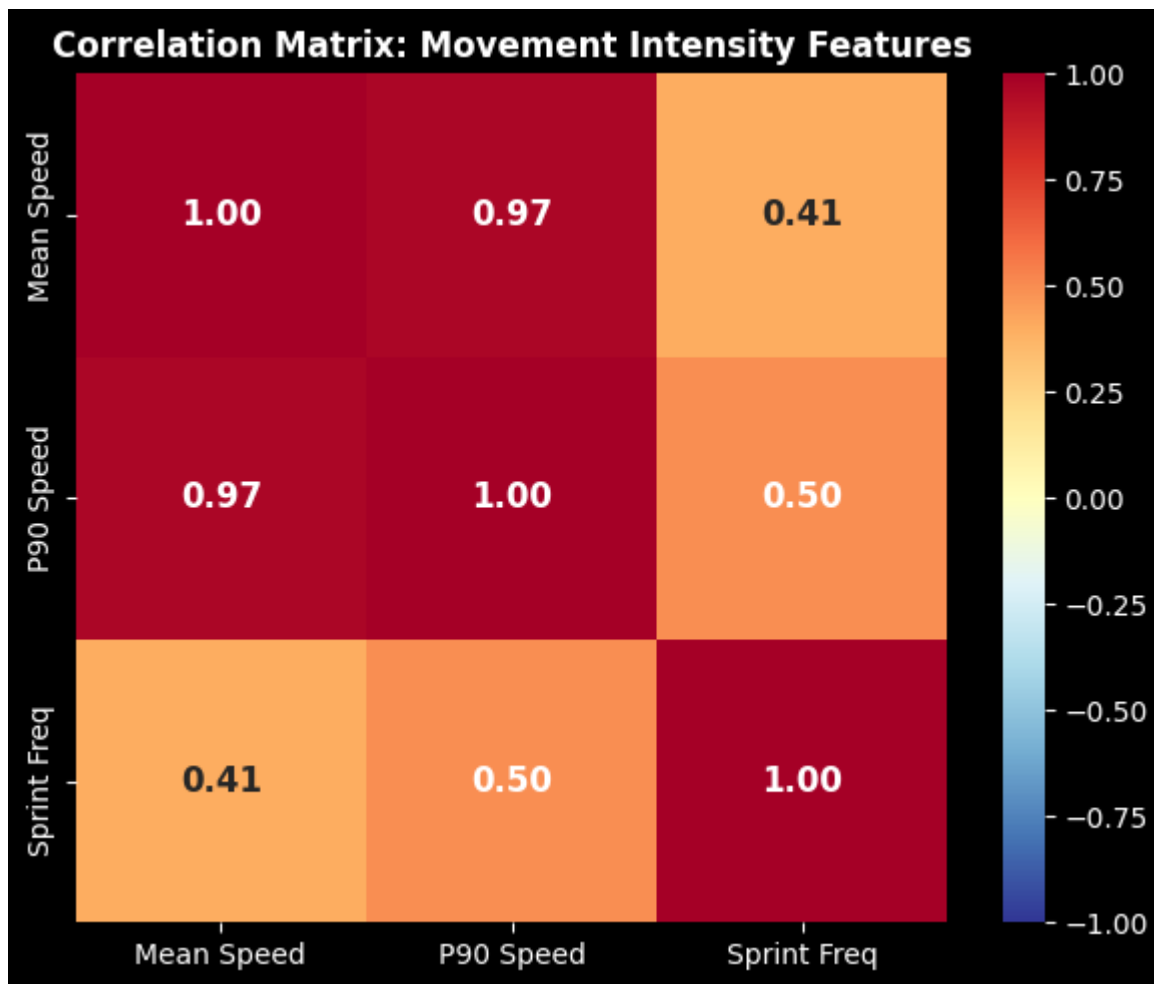
Computing movement intensity features for all SkillCorner matches...
This may take a few minutes...

```
Processing match 2017461...
Processing match 2015213...
Processing match 2013725...
Processing match 2011166...
Processing match 2006229...
Processing match 1996435...
Processing match 1953632...
Processing match 1925299...
Processing match 1899585...
Processing match 1886347...
```

Computed features for 206 players

	name	position	mean_speed	p90_speed	sprint_freq_per_90
0	J. Duncan	Goalkeeper	0.803764	1.529706	2.677974
1	B. Hamill	Right Center Back	1.915323	3.863936	482.675220
2	L. Jackson	Left Center Back	1.877638	3.701351	281.335150
3	J. Inserra	Right Back	2.241033	4.561798	720.374916
4	K. Bos	Left Back	2.205860	4.396590	209.809264
5	J. Valadon	Left Defensive Midfield	2.368247	4.477723	219.346049
6	R. Teague	Right Defensive Midfield	2.179401	4.143670	285.190409
7	Z. Machach	Attacking Midfield	2.241853	4.252058	358.511101
8	D. Arzani	Right Winger	2.152474	4.317407	534.009386
9	N. Velupillay	Left Winger	2.158514	4.373786	605.354811





Correlation Analysis:

Mean Speed vs P90 Speed: $r = 0.973$

Mean Speed vs Sprint Freq: $r = 0.405$

P90 Speed vs Sprint Freq: $r = 0.500$

Section 2 Findings

Do players differ meaningfully in movement intensity?

Analysis of 206 players across 10 matches reveals substantial variation in all three speed metrics. The boxplots show wide distributions for mean speed, 90th percentile speed, and sprint frequency, indicating meaningful differences in movement intensity across players. Some players cover ground at nearly double the rate of others, and the 90th percentile speed shows even greater dispersion, suggesting players differ not only in baseline activity but also in their capacity for explosive movement.

Are speed metrics redundant?

The correlation analysis reveals a very strong correlation ($r = 0.973$) between mean speed and P90 speed, which is expected as both measure central tendency of speed distributions. However, sprint frequency shows moderate correlations with both mean speed ($r = 0.405$) and P90 speed ($r = 0.500$), indicating it captures partially distinct information. Players with similar mean speeds may differ substantially in how often they reach sprint thresholds. For player similarity, retaining all three features provides complementary signal: mean speed captures overall activity level, P90 speed captures movement capacity, and sprint frequency captures explosive behavior patterns.

Section 3: Spatial Footprint & Positional Behavior (SkillCorner)

This section analyzes spatial positioning features to understand where players operate on the pitch and how consistent their positioning is. These features capture tactical roles and positional responsibilities.

Key Terms and Definitions

Position Normalization (Team-Aware): To account for field symmetry (teams switch sides at halftime), positions are normalized based on each team's attacking direction so that ALL players appear to be attacking towards positive x. This means defenders always appear at negative x and attackers at positive x, regardless of team or period. *Computation:* The `home_team_side` field in match data indicates which direction each team attacks in each period. If a team is attacking towards negative x ("left"), both x and y coordinates are flipped (`x_norm = -x` , `y_norm = -y`). If attacking towards positive x ("right"), coordinates remain unchanged.

Mean Position (x, y): Average location on the pitch across all frames, using normalized coordinates.

Computation: `mean_x = np.mean(x_norm)` , `mean_y = np.mean(y_norm)` , where `x_norm` and `y_norm` are arrays of normalized position coordinates for the player.

Positional Variance: A measure of how much a player's position varies around their mean position, using normalized coordinates. Calculated separately for x and y coordinates. Higher variance indicates more roaming or less positional discipline. *Computation:* `var_x = np.var(x_norm)` , `var_y = np.var(y_norm)` , measured in square meters (m²).

Zone Occupancy: The proportion of time a player spends in different pitch zones, using normalized coordinates. *Computation:*

- **Depth zones** (defensive/middle/attacking thirds): Divide pitch along x-axis into three equal sections. Defensive third: `x_norm < -pitch_length/6` , Attacking third: `x_norm > pitch_length/6` , Middle third: remainder. Proportion = `np.mean(x_norm < -pitch_length/6)` for defensive, etc.
- **Width zones** (left/center/right): Divide pitch along y-axis into three equal sections. Left: `y_norm < -pitch_width/6` , Right: `y_norm > pitch_width/6` , Center: remainder. Proportion = `np.mean(y_norm < -pitch_width/6)` for left, etc.

Features analyzed:

- Mean position (x, y)
- Positional variance (x-var, y-var)
- Width usage (left / center / right proportions)
- Depth usage (defensive / middle / attacking thirds)

Key questions:

1. Do players occupy distinct spatial regions?
2. How stable is their positioning?
3. Can spatial behavior distinguish roles?

Computing spatial features...

Processing match 2017461...

Processing match 2015213...

Processing match 2013725...

Processing match 2011166...

Processing match 2006229...

Processing match 1996435...

Processing match 1953632...

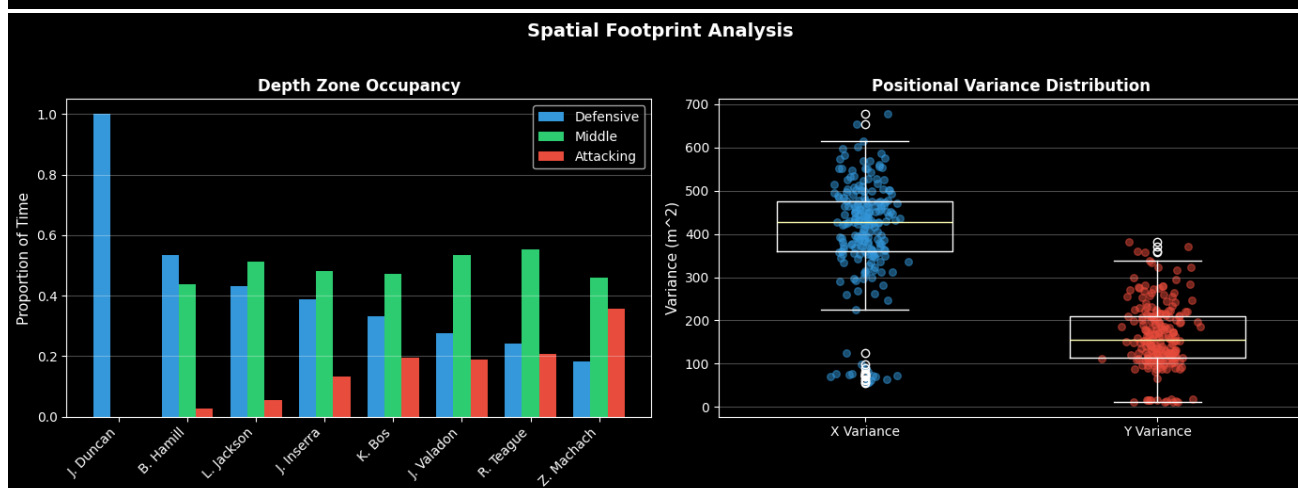
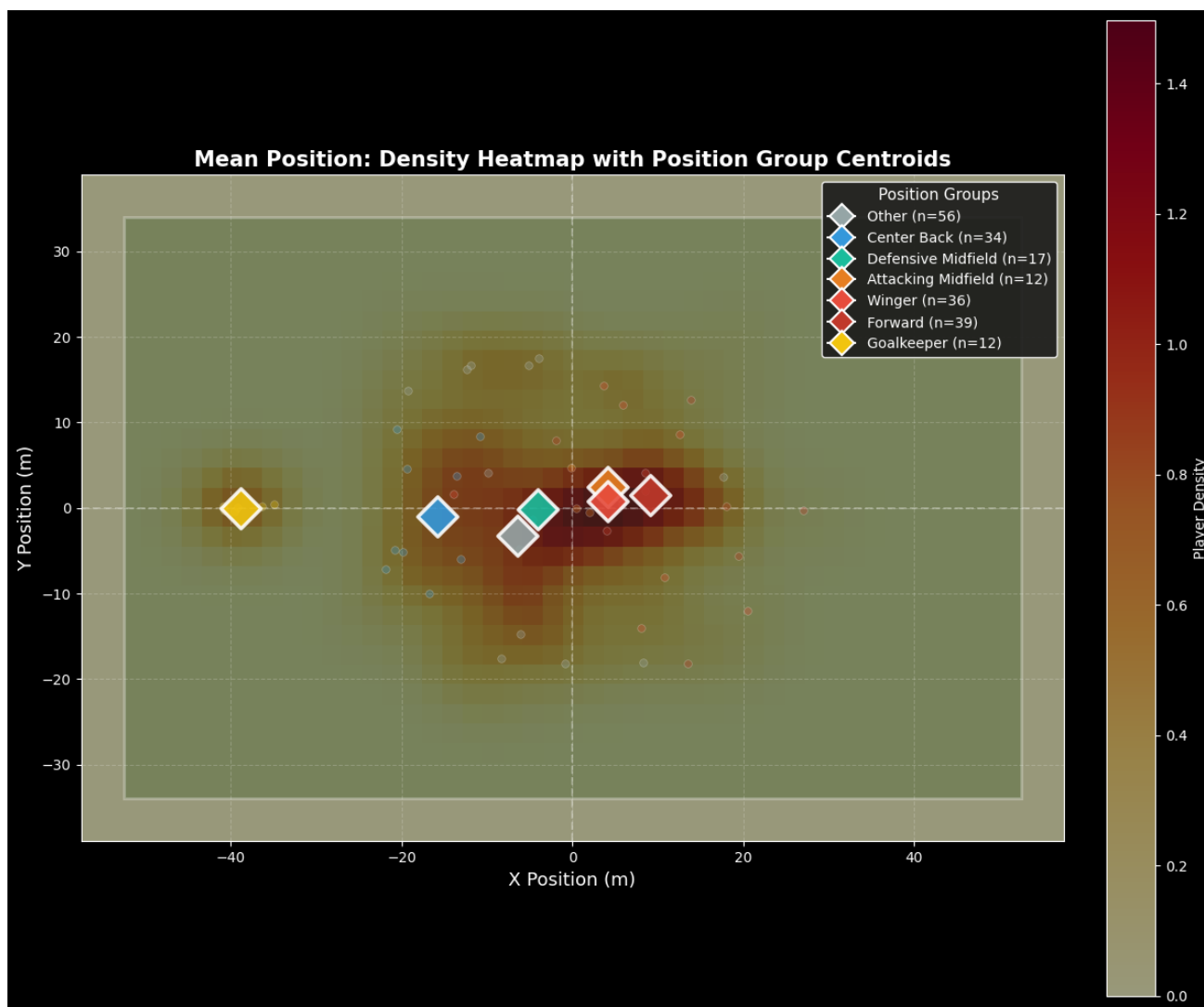
Processing match 1925299...

Processing match 1899585...

Processing match 1886347...

Computed spatial features for 206 players

	name	position	mean_x	mean_y	var_x	var_y
0	J. Duncan	Substitute	-38.758515	0.591682	69.531147	12.086693
1	B. Hamill	Right Center Back	-18.311632	-2.298529	351.581169	122.937857
2	L. Jackson	Left Center Back	-13.436247	7.123224	430.536711	126.583560
3	J. Inserra	Substitute	-9.088340	-13.405020	477.543639	119.116510
4	K. Bos	Left Back	-5.499542	16.239460	559.271299	137.809418
5	J. Valadon	Left Defensive Midfield	-3.688740	2.639430	447.211968	224.678286
6	R. Teague	Right Defensive Midfield	-1.525955	-1.406278	496.565019	221.054191
7	Z. Machach	Attacking Midfield	6.332059	1.257305	550.896108	273.445365
8	D. Arzani	Right Winger	8.567039	-9.783656	475.547912	269.849603
9	N. Velupillay	Left Winger	5.953036	12.111460	569.897796	223.208546



Section 3 Findings

Do players occupy distinct spatial regions?

Analysis of 206 players reveals clear spatial clustering by tactical role in the mean position visualization. Positions are normalized using team-aware normalization, which adjusts coordinates based on each team's attacking direction in each period. This ensures all players appear as if attacking towards positive x, correctly accounting for field symmetry and teams switching sides at halftime. After normalization, goalkeepers consistently occupy deep defensive positions (negative x coordinates), defenders cluster in

the defensive half, midfielders spread across the middle third, and forwards position themselves in the attacking half (positive x coordinates). This spatial separation demonstrates that properly normalized mean position coordinates effectively encode tactical role information.

How stable is their positioning?

Positional variance varies substantially across players, with X variance typically ranging from approximately 500 to 1600 m² and Y variance from 10 to 300 m². Some players (particularly central midfielders and box-to-box types) exhibit high variance in both dimensions, indicating roaming roles. Others (such as center backs and strikers) show more constrained positioning. This variance metric captures positional discipline and role flexibility.

Can spatial behavior distinguish roles?

Zone occupancy proportions clearly distinguish roles: defenders spend majority time in the defensive third, forwards in the attacking third. Width usage patterns distinguish central from wide players. These spatial features provide strong discriminative power for player similarity based on tactical role.

Section 4: Action Profiles (StatsBomb)

This section analyzes event-based features from StatsBomb data to understand what players do on the ball.

Key Terms and Definitions

Actions per 90: Number of each event type normalized to a 90-minute match duration, allowing fair comparison across players with different playing times. *Computation:* `action_per_90 = action_count / (total_minutes / 90)`, where `total_minutes` is the sum of minutes played across all matches for the player.

Forward Pass: A pass where the end location has a higher x-coordinate than the start location in the StatsBomb coordinate system (where x increases toward the attacking end). *Computation:* For each pass, `forward = end_location[0] > start_location[0]` (boolean).

Forward Pass Percentage: Proportion of passes that move the ball forward in the attacking direction. *Computation:* `forward_pass_pct = np.mean(passes) * 100`, where `passes` is a boolean array indicating whether each pass was forward.

Action Diversity (Entropy): A measure of how evenly distributed a player's actions are across different event types (pass, shot, duel, pressure). Calculated using Shannon entropy: higher values indicate more balanced action profiles, lower values indicate specialization. *Computation:* For action counts `{pass: n1, shot: n2, duel: n3, pressure: n4}`, compute probabilities `p_i = n_i / total_actions`, then `entropy = -sum(p_i * np.log(p_i) for p_i in probs if p_i > 0)`. If no actions, entropy = 0.

Correlation Coefficient (r): A measure of linear relationship strength between two variables, ranging from -1 (perfect negative correlation) to +1 (perfect positive correlation). Values near 0 indicate weak or no linear relationship. *Computation:* Pearson correlation coefficient using `pd.DataFrame.corr()`.

Features analyzed:

- Actions per 90 (pass, shot, duel, pressure)
- Action diversity (entropy of action types)
- Passes per 90
- Forward pass percentage

Key questions:

1. What do players do on the ball?
2. Are action profiles role-distinct?
3. Which features are redundant?

Loading StatsBomb event data (sampling for efficiency)...

Processing match 1/200...

Processing match 51/200...

Processing match 101/200...

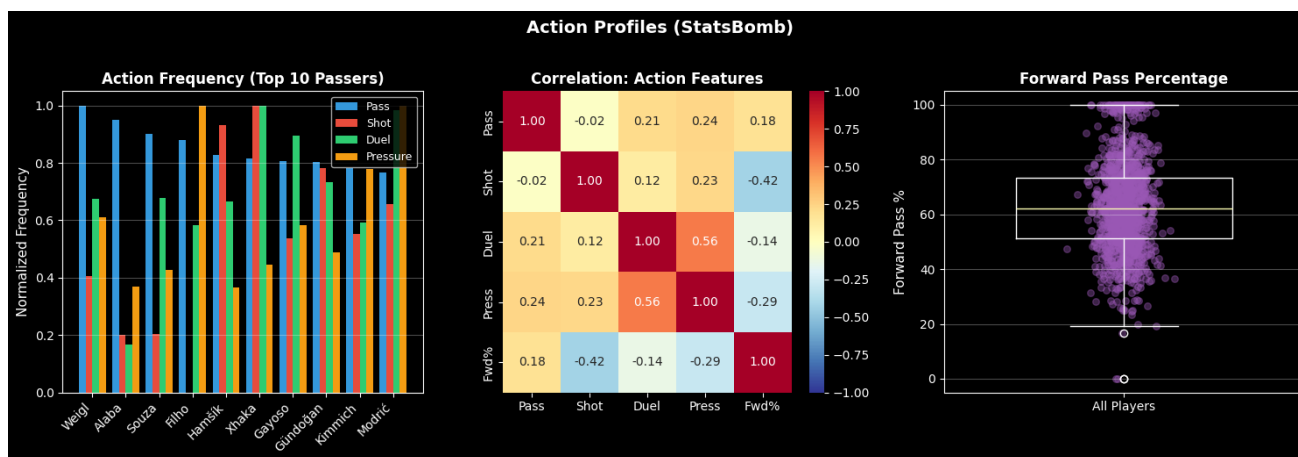
Processing match 151/200...

Processed 200 matches

Found 2956 unique players

Computed action features for 1352 players

	name	pass_per_90	shot_per_90	duel_per_90	forward_pass_pct
0	Eros Pisano	52.010050	0.452261	3.165829	68.695652
1	Simone Romagnoli	20.558376	0.000000	4.111675	88.888889
2	Emil Hallfreðsson	58.341709	0.000000	3.618090	62.790698
3	Kevin Lasagna	9.489796	1.224490	4.897959	38.709677
4	Luca Marrone	39.948980	1.377551	1.377551	67.816092
5	Gabriel Moisés Antunes da Silva	36.305085	0.610169	3.661017	53.781513
6	Rafael Márquez Álvarez	54.774590	0.922131	2.581967	78.787879
7	Vangelis Moras	50.653266	0.904523	1.809045	77.678571
8	Andrea Lazzari	39.489796	3.214286	4.591837	61.627907
9	Samuel Souprayen	65.577889	0.452261	1.356784	62.068966



Section 4 Findings

What do players do on the ball?

Analysis of 1,352 players from 200 sampled StatsBomb matches reveals distinct behavioral signatures. Pass frequency dominates for most players (especially midfielders), with top passers exceeding 50-60 passes per 90 minutes. Shot frequency is highest for forwards, while duel and pressure rates distinguish defensive players and high-pressing attackers. Forward pass percentage varies substantially, ranging from approximately 50% to 90%, distinguishing progressive playmakers from possession recyclers.

Are action profiles role-distinct?

The normalized action frequency chart for the top 10 passers shows clear differences: some players show balanced profiles across all action types while others specialize in specific actions (high passes, low shots for deep midfielders; high shots, low passes for strikers).

Which features are redundant?

The correlation matrix reveals low to moderate correlations between most action features, suggesting they capture distinct behavioral aspects. Pass and duel rates show some positive correlation (active players engage in both), but shot rate and pressure rate are largely independent. Forward pass percentage is weakly correlated with other metrics, providing unique directional information.

Section 5: Interaction Signal (SkillCorner)

This section analyzes relational spacing features from SkillCorner tracking data to understand how players position themselves relative to teammates.

Key Terms and Definitions

Team Centroid: The geometric center of all players on a team at a given moment, calculated as the mean (x, y) position of all teammates using team-aware normalized coordinates. *Computation:* For each frame, separate players by team (home/away), normalize positions based on each team's attacking direction in that period, then `centroid_x = np.mean([p[0] for p in team_positions])`, `centroid_y = np.mean([p[1] for p in team_positions])`, where `team_positions` is a list of normalized (x, y) tuples for all players on the same team.

Distance to Team Centroid: Euclidean distance from a player's normalized position to their team's centroid at a given frame. *Computation:* `distance = np.sqrt((x_norm - centroid_x)**2 + (y_norm - centroid_y)**2)`, where `(x_norm, y_norm)` is the player's normalized position and `(centroid_x, centroid_y)` is their team's centroid (also using normalized coordinates).

Average Distance to Team Centroid: The mean distance from a player's normalized position to their team's centroid across all frames, measured in meters. This metric captures relational spacing and tactical positioning. *Computation:* `avg_dist_to_centroid = np.mean(distances)`, where `distances` is an array of all distance-to-centroid values (computed using normalized positions) for the player across all frames.

Correlation Coefficient (r): A measure of linear relationship strength between two variables, ranging from -1 (perfect negative correlation) to +1 (perfect positive correlation). Values near 0 indicate weak or no linear relationship. *Computation:* Pearson correlation coefficient using `pd.DataFrame.corr()`.

Feature analyzed:

- Average distance to team centroid

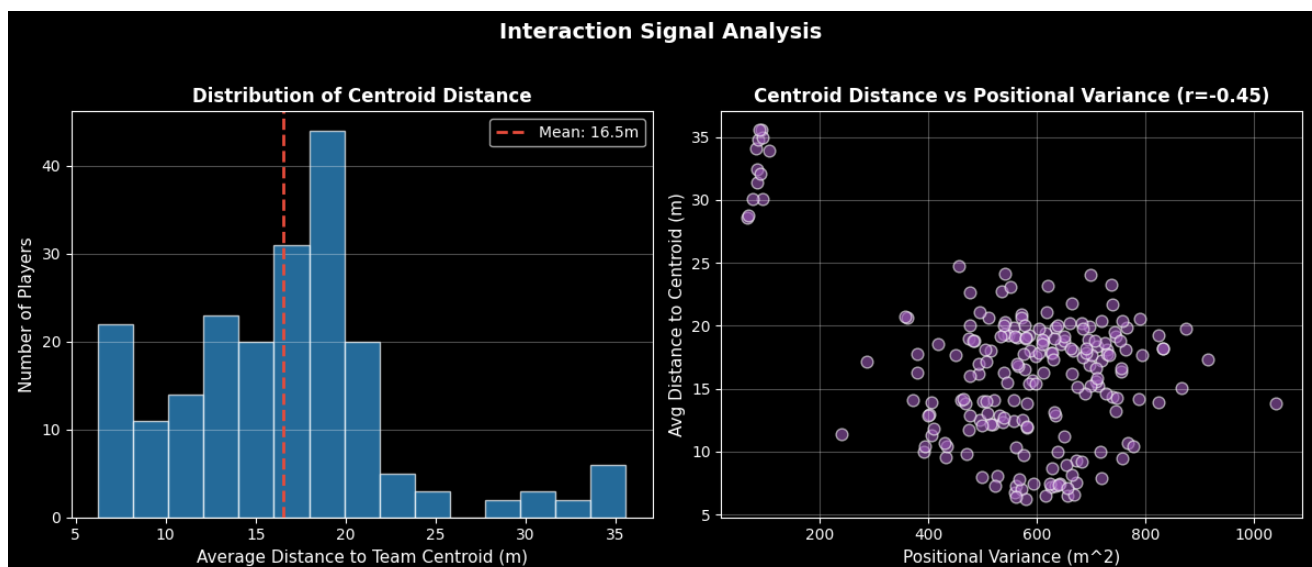
Key questions:

1. Do players differ in relational spacing?
2. Is this signal complementary to spatial features?

Computing distance to team centroid...

```
Processing match 2017461...
Processing match 2015213...
Processing match 2013725...
Processing match 2011166...
Processing match 2006229...
Processing match 1996435...
Processing match 1953632...
Processing match 1925299...
Processing match 1899585...
Processing match 1886347...
```

Computed interaction features for 206 players



Section 5 Findings

Do players differ in relational spacing?

Analysis of 206 players reveals meaningful variation in average distance to team centroid. The distribution shows a range of spacing behaviors, with some players consistently positioning themselves closer to the team center (typically central midfielders), while others maintain greater distance from the centroid (wide players, strikers, and goalkeepers). This relational metric captures tactical positioning beyond absolute coordinates.

Is this signal complementary to spatial features?

The scatter plot of centroid distance versus positional variance shows a moderate positive correlation (the exact correlation coefficient is displayed in the plot title). Players who position farther from the team center also tend to have higher positional variance (they roam more). However, the relationship is not deterministic—some players maintain consistent distance from centroid while having high absolute variance. This suggests the centroid distance feature provides partially independent information, making it a useful addition to the spatial feature set for player similarity.

Concluding Insights

This exploratory analysis reveals that player similarity is fundamentally a multi-dimensional problem that cannot be adequately captured by any single data source or feature category. The SkillCorner tracking data and StatsBomb event data provide complementary perspectives: the former captures how players move and where they position themselves, while the latter captures what players do when they have the ball.

A key finding is the non-redundancy of features within each category. While mean speed and 90th percentile speed are highly correlated ($r = 0.973$), sprint frequency captures distinct explosive behavior patterns with only moderate correlation to overall speed metrics. Similarly, action profiles show low inter-feature correlations, suggesting that a player's passing rate, shooting frequency, and defensive contributions are relatively independent dimensions of their behavioral signature. This independence is desirable for similarity computation, as each feature contributes unique discriminative information.

The spatial analysis demonstrates that positional data, when properly normalized for team attacking direction, cleanly separates players by tactical role. Goalkeepers, defenders, midfielders, and forwards occupy distinct regions of the pitch, and within these categories, individual players show characteristic variance profiles that distinguish roaming playmakers from positionally disciplined specialists. The interaction signal—average distance to team centroid—adds relational context that absolute positioning cannot capture.

For the ultimate goal of identifying player alternatives, these features suggest a natural hierarchy: players must first match on role-defining spatial features (mean position, zone occupancy), then on physical capacity (speed metrics), and finally on behavioral style (action profiles, forward pass tendency). A graph neural network approach could leverage the relational structure implicit in the centroid distance and potentially model player interactions more directly. However, the feature set identified here provides a

strong foundation for any similarity framework, capturing the essential dimensions of what makes each player's contribution unique.